

SUGGESTED SOLUTION SET A

Section A

1.

(a)

$$\begin{aligned}\lim_{x \rightarrow 1} f(x) - \lim_{x \rightarrow 0} f(x) &= \lim_{x \rightarrow 1} \frac{x^2 - 1}{x - 1} - \lim_{x \rightarrow 0} \frac{x^2 - 1}{x - 1} \\&= \lim_{x \rightarrow 1} \frac{(x+1)(x-1)}{(x-1)} - \frac{0-1}{0-1} \\&= \lim_{x \rightarrow 1} (x+1) - (1) \\&= (1+1) - 1 \\&= 1\end{aligned}$$

(b)

$$\begin{aligned}\lim_{x \rightarrow +\infty} \frac{\sqrt{x^2 - 1}}{3x + 5} &= \lim_{x \rightarrow +\infty} \frac{\frac{\sqrt{x^2 - 1}}{\sqrt{x^2}}}{\frac{3x + 5}{x}} \\&= \lim_{x \rightarrow +\infty} \frac{\sqrt{1 - \frac{1}{x^2}}}{3 - \frac{5}{x}} \\&= \frac{\sqrt{1 - 0}}{3 - 0} \\&= \frac{1}{3}\end{aligned}$$

2. (a) $y = x \ln x - 3x$

$$\frac{dy}{dx} = x \left(\frac{1}{x} \right) + (\ln x)(1) - 3$$

$$\frac{dy}{dx} = \ln x - 2$$

$$0 = \ln x - 2$$

$$\ln x = 2$$

$$x = e^2$$

$$y = e^2 \ln e^2 - 3e^2$$

$$y = 2e^2 - 3e^2$$

$$y = -e^2$$

(b) $y = \alpha e^{2x} + \beta e^{-2x}$

$$\frac{dy}{dx} = \alpha (2e^{2x}) + \beta (-2e^{-2x})$$

$$\frac{dy}{dx} = 2\alpha e^{2x} - 2\beta e^{-2x}$$

$$\frac{d^2y}{dx^2} = 2\alpha (2e^{2x}) - 2\beta (-2e^{-2x})$$

$$\frac{d^2y}{dx^2} = 4\alpha e^{2x} + 4\beta e^{-2x}$$

$$\frac{d^2y}{dx^2} = 4(\alpha e^{2x} + \beta e^{-2x})$$

$$\frac{d^2y}{dx^2} = 4y \quad \text{shown}$$

$$3. \quad f(x) = x^2 - \frac{2}{x}$$

$$f'(x) = 2x + \frac{2}{x^2}$$

$$0 = 2x + \frac{2}{x^2}$$

$$x^3 = -1$$

$$x = -1$$

$$\text{When } x = -1, \quad f(-1) = (-1)^2 - \frac{2}{(-1)} = 3$$

the coordinate of stationary point $(-1, 3)$

$$f'(x) = 2x + \frac{2}{x^2}$$

$$f''(x) = 2 - \frac{4}{x^3}$$

$$f''(-1) = 2 - \frac{4}{(-1)^3} = 6 > 0$$

(relative minimum)

Section B

$$1. (a) \quad z_1 + z_2 = 2 + 3i + 3 + 2i$$

$$= 5 + 5i$$

$$|z_1 + z_2| = \sqrt{5^2 + 5^2}$$

$$= \sqrt{50}$$

$$= 5\sqrt{2}$$

$$\begin{aligned}
 \text{(b)} \quad z &= \frac{z_1 z_3}{z_2} = \frac{(2+3i)(a+bi)}{(3+2i)} \times \frac{(3-2i)}{(3-2i)} \\
 &= \frac{(6-4i+9i-6i^2)(a+bi)}{(3+2i)(3-2i)} \\
 &= \frac{(12+5i)(a+bi)}{9+4} \\
 &= \frac{12a+12bi+5ai+5bi^2}{13} = \frac{12a-5b}{13} + \frac{(5a+12b)}{13}i
 \end{aligned}$$

$$\text{(c)(i)} \quad z = \frac{17}{13} - \frac{7}{13}i$$

$$\therefore \frac{17}{13} - \frac{7}{13}i = \frac{12a-5b}{13} + \frac{(5a+12b)}{13}i$$

$$\frac{12a-5b}{13} = \frac{17}{13}$$

$$\frac{5a+12b}{13} = -\frac{7}{13}$$

$$12a-5b=17 \text{ --- (1)}$$

$$5a+12b=-7 \text{ --- (2)}$$

$$(1) \times 5, \quad 60a-25b=85 \text{ --- (3)}$$

$$(2) \times 12, \quad 60a+144b=-84 \text{ --- (4)}$$

$$(4) - (3), \quad 169b = -169$$

$$\therefore b = -1, \quad \therefore a = 1$$

$$\text{(ii)} \quad \tan \theta = \frac{-\frac{7}{13}}{\frac{17}{13}}$$

$$\theta = \tan^{-1} \left| \frac{-7}{17} \right| = 0.3906 \text{ rad}$$

$$\arg z = -0.391 \text{ rad (3d.p)}$$

$$\begin{aligned}
 \text{(iii)} \quad |z| &= \sqrt{\left(\frac{17}{13}\right)^2 + \left(\frac{-7}{13}\right)^2} \\
 &= \sqrt{\frac{338}{169}} = \sqrt{2}
 \end{aligned}$$

The polar form

$$z = \sqrt{2} (\cos(-0.391 \text{ rad}) + i \sin(-0.391 \text{ rad}))$$

$$2.(a) \quad \log_5 5x + \log_x 5 = \log_{25} 15625$$

$$\log_5 5 + \log_5 x + \frac{1}{\log_5 x} = \frac{\log_5 15625}{\log_5 25}$$

$$1 + \log_5 x + \frac{1}{\log_5 x} = \frac{\log_5 5^6}{\log_5 5^2}$$

$$1 + \log_5 x + \frac{1}{\log_5 x} = \frac{6 \log_5 5}{2 \log_5 5}$$

$$1 + \log_5 x + \frac{1}{\log_5 x} = 3$$

$$\log_5 x + \frac{1}{\log_5 x} = 2$$

$$\text{Let } a = \log_5 x, \quad a + \frac{1}{a} = 2$$

$$a^2 - 2a + 1 = 0$$

$$(a-1)^2 = 0$$

$$\therefore a = 1$$

$$\therefore \log_5 x = 1$$

$$\therefore x = 5$$

(b) $3 - x \leq x(x - 3) < 10 - 3x$

$3 - x \leq x(x - 3)$ and

$x(x - 3) < 10 - 3x$

$3 - x \leq x^2 - 3x$

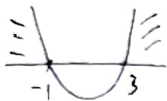
$x^2 - 3x < 10 - 3x$

$x^2 - 2x - 3 \geq 0$

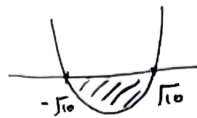
$x^2 - 10 < 0$

$(x + 1)(x - 3) \geq 0$

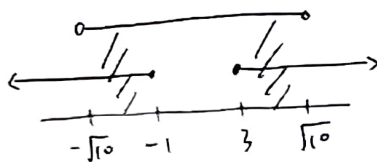
$(x + \sqrt{10})(x - \sqrt{10}) < 0$



$\therefore (-\infty, -1] \cup [3, \infty)$



$\therefore (-\sqrt{10}, \sqrt{10})$



$\therefore (-\sqrt{10}, -1] \cup [3, \sqrt{10})$

3. (a)

$f(x) = \frac{x+2}{x-2}, x \neq k$

$x - 2 \neq 0$

$x \neq 2$

$k = 2$

b)

$f(x_1) = f(x_2)$

$\frac{x_1 + 2}{x_1 - 2} = \frac{x_2 + 2}{x_2 - 2}$

$x_1 x_2 - 2x_1 + 2x_2 - 4 = x_1 x_2 + 2x_1 - 2x_2 - 4$

$4x_1 = 4x_2$

$x_1 = x_2$

$\therefore f$ is a 1-1 function.

(c)

$$\begin{aligned}f[g(x)] &= f\left(\frac{2x+2}{x-1}\right) \\&= \frac{\frac{2x+2}{x-1} + 2}{\frac{2x+2}{x-1} - 2} \\&= \frac{2x+2+2x-2}{2x+2-2x+2} \\&= \frac{x-1}{x-1} \\&= \frac{4x}{4} \\&= x\end{aligned}$$

$\therefore f[g(x)] = x, \therefore f$ and g are inverse each other.

4. (a)

$$\begin{aligned}f(x) &= \ln(3x-1) \\f(f^{-1}(x)) &= \ln(3f^{-1}(x)-1) \\x &= \ln(3f^{-1}(x)-1) \\e^x &= 3f^{-1}(x)-1 \\f^{-1}(x) &= \frac{e^x+1}{3}\end{aligned}$$

(b)

$$D_{f^{-1}} = R_f = (-\infty, \infty) ; \quad R_{f^{-1}} = D_f = \left(\frac{1}{3}, \infty\right)$$

(d)

$$\begin{aligned}g(f(x)) &= g(\ln(3x-1)) \\&= e^{\ln(3x-1)} \\&= 3x-1\end{aligned}$$

5.

$$f(2) = p$$

$$8 - p = p$$

$$2p = 8$$

$$p = 4$$

$$f(2) = \lim_{x \rightarrow 2^-} r(x+6)$$

$$p = r(2+6)$$

$$4 = 8r$$

$$r = \frac{1}{2}$$

$$f(q) = \lim_{x \rightarrow q^-} \frac{(2-x)^2 - 4}{x}$$

$$\frac{1}{2}(q+6) = \frac{(2-q)^2 - 4}{q}$$

$$\frac{q+6}{2} = \frac{q^2 - 4q}{q}$$

$$q^2 + 6q = 2q^2 - 8q$$

$$q^2 - 14q = 0$$

$$q = 0 \text{ or } q = 14$$

$$\therefore q \leq x < 2, \therefore q = 0$$

6. (a) $x = 2; y^2 + 6(2) = 3(2)y + y$

$$y^2 - 7y + 12 = 0$$

$$(y-4)(y-3) = 0$$

$$y = 4 \text{ or } y = 3$$

$\therefore$ Points are $(2, 4)$ and $(2, 3)$

$$y^2 + 6x = 3xy + y$$

$$2y \frac{dy}{dx} + 6 = 3x \frac{dy}{dx} + y(3) + \frac{dy}{dx}$$

$$2y \frac{dy}{dx} - 3x \frac{dy}{dx} - \frac{dy}{dx} = 3y - 6$$

$$\frac{dy}{dx}(2y - 3x - 1) = 3y - 6$$

$$\frac{dy}{dx} = \frac{3y - 6}{2y - 3x - 1}$$

$$(2, 4): \frac{dy}{dx} = \frac{3(4) - 6}{2(4) - 3(2) - 1} = \frac{6}{1} = 6$$

$$(2, 3): \frac{dy}{dx} = \frac{3(3) - 6}{2(3) - 3(2) - 1} = \frac{3}{-1} = -3$$

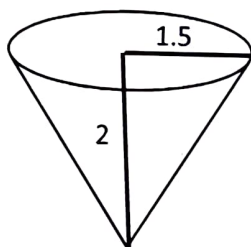
$$(b) \quad x = \frac{1}{3} \sin^3 2\theta \quad ; \quad y = 2 \cos^3 2\theta$$

$$\begin{aligned} \frac{dx}{d\theta} &= \frac{1}{3} \sin^2 2\theta (2 \cos 2\theta) \quad ; \quad \frac{dy}{d\theta} = 6 \cos^2 2\theta (-2 \sin 2\theta) \\ &= 2 \sin^2 2\theta \cos 2\theta \quad ; \quad = -12 \cos^2 2\theta \sin 2\theta \end{aligned}$$

$$\frac{dy}{dx} = \frac{dy}{d\theta} \times \frac{d\theta}{dx} = \frac{-12 \cos^2 2\theta \sin 2\theta}{2 \sin^2 2\theta \cos 2\theta} = -6 \cot 2\theta$$

$$\theta = \frac{\pi}{6} ; \quad \frac{dy}{dx} = -6 \cot 2\theta = \frac{-6}{\tan 2\theta} = \frac{-6}{\tan 2\left(\frac{\pi}{6}\right)} = \frac{-6}{\sqrt{3}} = -2\sqrt{3}$$

7.



$$\frac{r}{h} = \frac{1.5}{2}$$

$$r = \frac{3}{4}h$$

$$\frac{dV}{dt} = -0.25 m^3 / \text{min}$$

$$V = \frac{1}{3} \pi r^2 h = \frac{1}{3} \pi \left(\frac{3}{4}\right)^2 h = \frac{3}{16} \pi h^3$$

$$\frac{dV}{dh} = \frac{9}{16} \pi h^2$$

$$\begin{aligned} (a) \quad \frac{dh}{dt} &= \frac{dV}{dt} \times \frac{dh}{dV} \\ &= -0.25 \times \frac{16}{9\pi(1)^2} \\ &= -0.1415 m / \text{min} \end{aligned}$$

$$(b) \quad r = \frac{3}{4}h$$

$$\frac{dr}{dh} = \frac{3}{4}$$

$$\begin{aligned} \frac{dr}{dt} &= \frac{dr}{dh} \times \frac{dh}{dt} \\ &= -0.1415 \times \frac{3}{4} \\ &= -0.1061 m / \text{min} \end{aligned}$$